Supporting Information

Mendelian randomization and multi-omic evidence for regulatory peak–gene links in human immune cells

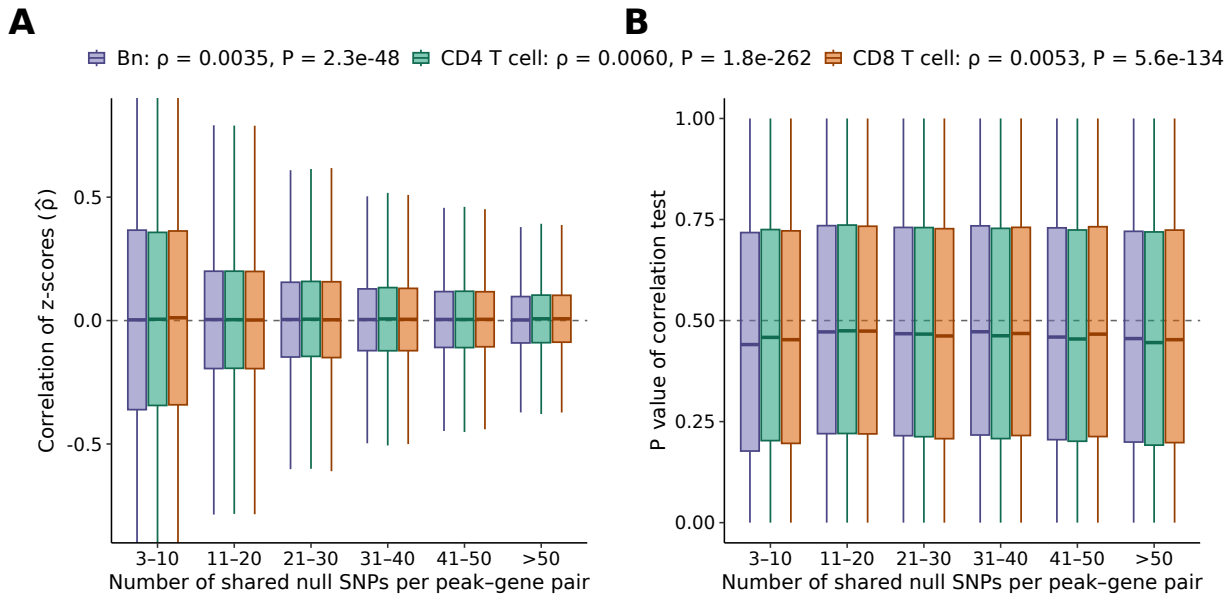
S1–S3 Figs and S1–S3 Tables accompany the main manuscript (`main.tex`).

Supporting figures

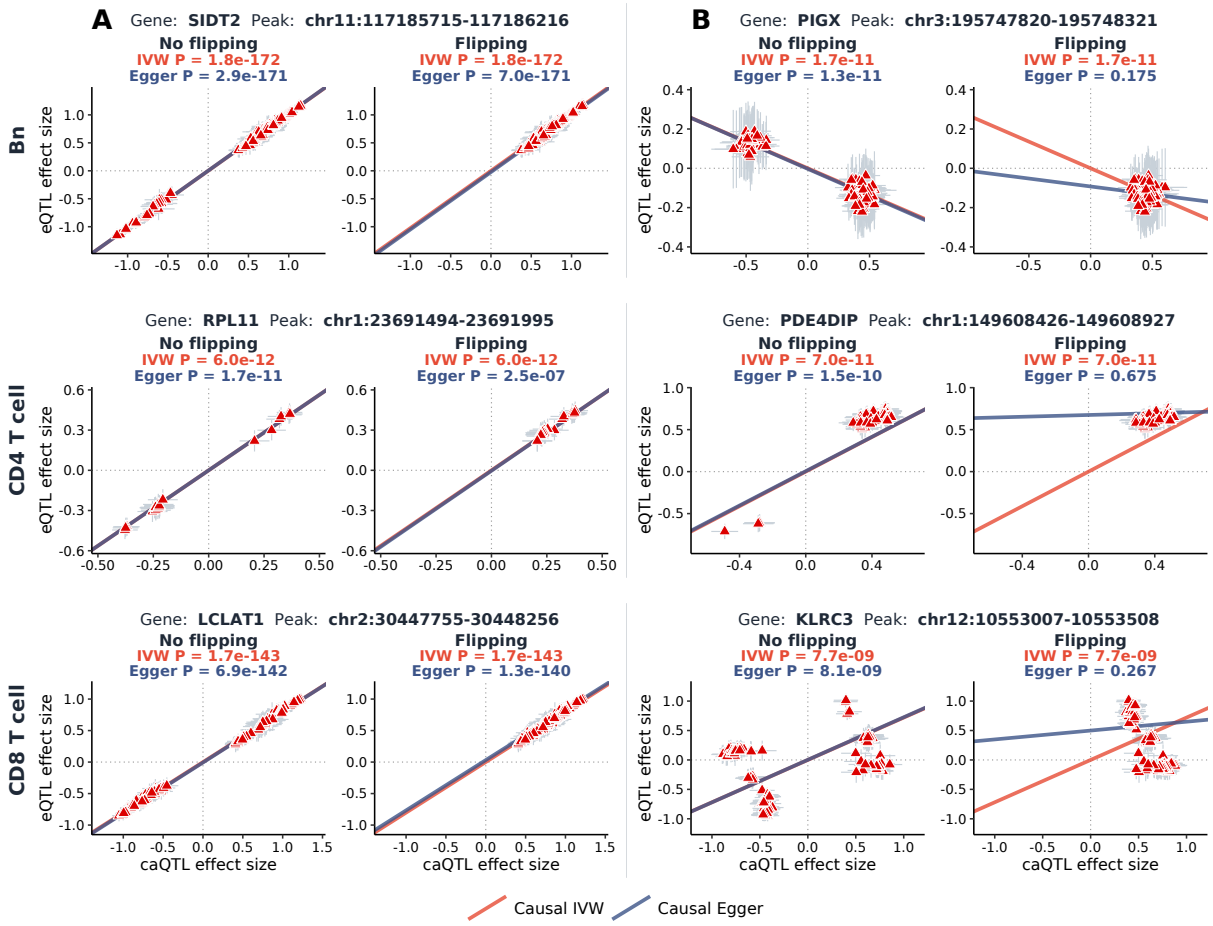
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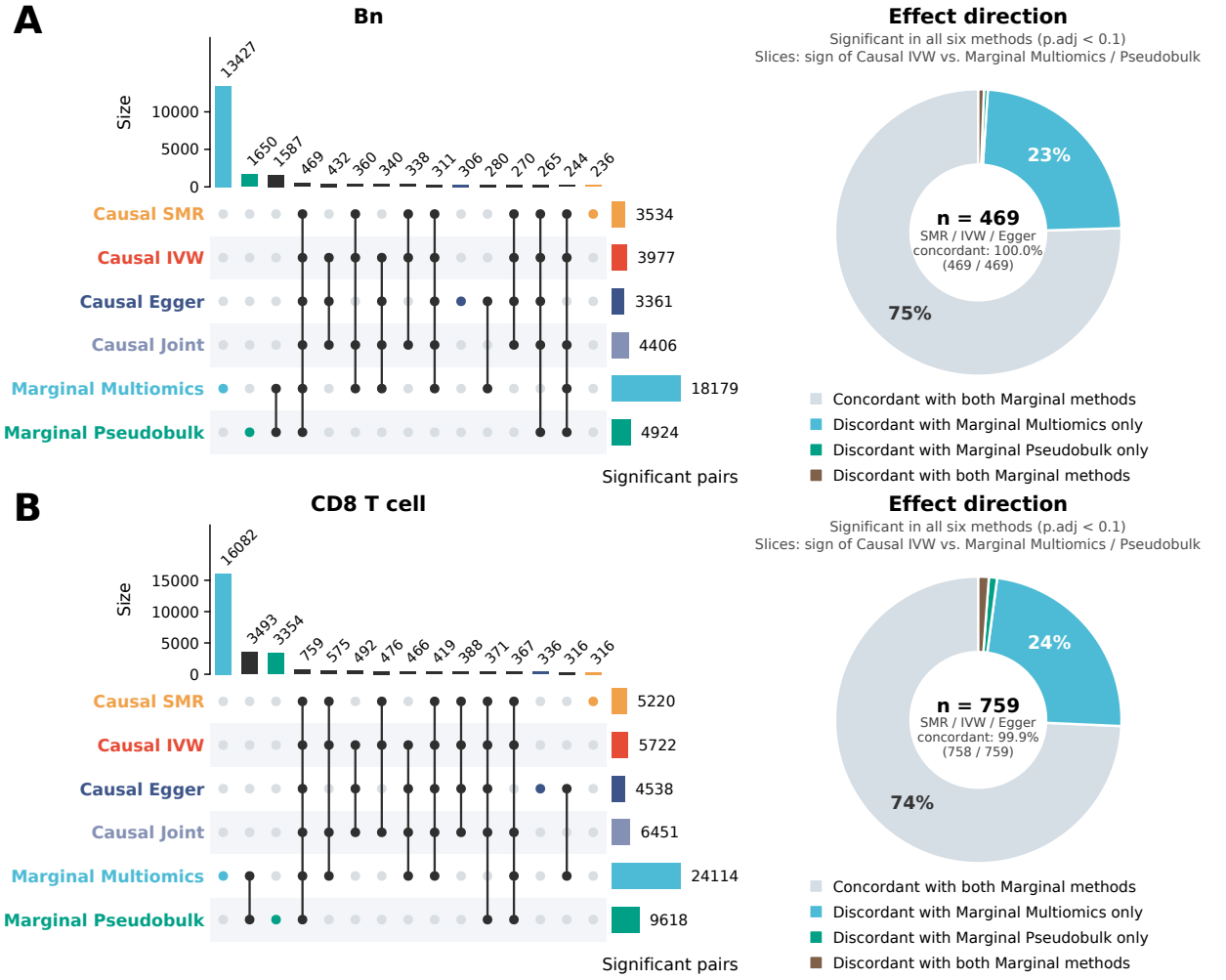
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S1 Fig. Null-SNP diagnostics in Bn, CD4 T cells, and CD8 T cells. **A,** Pair-level correlations of caQTL and eQTL z-scores, grouped by the number of shared pruned null SNPs; boxes show the three cell types side by side (Bn, purple; CD4 T cells, green; CD8 T cells, orange). The legend gives the global correlation ρ and its p-value for each cell type. **B,** Corresponding p-values of the pair-level correlation tests. The dashed lines mark zero correlation in A and a p-value of 0.5 in B, the values expected under the null. The analyses included 547,452 (Bn), 1,051,953 (CD4 T cells), and 686,829 (CD8 T cells) peak–gene pairs, based on 17,074,938, 33,263,695, and 21,537,822 SNP–pair observations, respectively. Global summaries pool SNP–pair observations, so a variant may occur in multiple pairs. This diagnostic does not itself establish MR calibration.



S2 Fig. MR-Egger estimates with and without allele orientation in Bn, CD4 T cells, and CD8 T cells. Rows show Bn, CD4 T cells, and CD8 T cells, and each pair of plots is one peak–gene pair. **A,** pairs whose estimates change little with orientation (SIDT2 in Bn, 194 SNPs; RPL11 in CD4 T cells, 25 SNPs; LCLAT1 in CD8 T cells, 195 SNPs). **B,** pairs whose MR-Egger estimate changes markedly (PIGX in Bn, 136 SNPs; PDE4DIP in CD4 T cells, 74 SNPs; KLRC3 in CD8 T cells, 134 SNPs). For each pair, the left plot (No flipping) uses the original allele coding and the right plot (Flipping) orients the pair so that the caQTL effect is positive, as in the MR analysis; both plots share axis ranges. Red and blue lines are the IVW and MR-Egger fits, and the P values above each plot are the corresponding IVW and MR-Egger P values. Red triangles are SNPs and gray bars are 95% confidence intervals. IVW is unchanged by orientation.



S3 Fig. Overlap and effect directions of significant links in Bn and CD8 T cells. **A,** Bn; **B,** CD8 T cells. In each row, left: UpSet plot of the 15 largest distinct significance combinations across six methods at within-gene BH-adjusted $q < 0.1$; top bars give the size of each combination, and right bars give the number of significant pairs for each method. Right: among links significant in all six methods, the sign of the IVW estimate relative to the signs of the two marginal correlations (categories in the legend below the ring); the center reports the number of such links and the proportion whose SMR, IVW, and Egger estimates agree in sign. "Causal Joint" denotes CCT. The corresponding CD4 T cell comparison is shown in Fig 1B–C.

S1 Table. Pooled hypergeometric enrichment p-values across regulatory references.

Cell type	Method	EnhancerAtlas	FANTOM5	4DGenome	GTEx	HiChIP	PCHiC
Bn	Causal SMR	3.7×10^{-136}	3.9×10^{-100}	2.5×10^{-58}	2.3×10^{-82}	1.0×10^{-416}	3.4×10^{-58}
	Causal IVW	1.2×10^{-108}	1.5×10^{-88}	3.1×10^{-52}	1.6×10^{-71}	2.7×10^{-365}	2.6×10^{-46}
	Causal Egger	6.0×10^{-65}	8.5×10^{-68}	5.6×10^{-40}	4.2×10^{-38}	9.0×10^{-286}	1.1×10^{-31}
	Causal Joint	3.7×10^{-130}	1.1×10^{-99}	2.3×10^{-54}	4.2×10^{-77}	5.3×10^{-427}	9.3×10^{-57}
	Marginal Multiomics	6.1×10^{-138}	1.3×10^{-162}	5.1×10^{-61}	2.5×10^{-61}	2.1×10^{-879}	2.9×10^{-75}
	Marginal Pseudobulk	1.6×10^{-80}	2.6×10^{-94}	2.4×10^{-44}	8.4×10^{-55}	8.0×10^{-404}	4.0×10^{-54}
CD4 T cell	Causal SMR	3.5×10^{-302}	6.5×10^{-107}	9.4×10^{-50}	2.6×10^{-209}	1.3×10^{-911}	2.0×10^{-140}
	Causal IVW	1.2×10^{-221}	3.0×10^{-94}	3.2×10^{-47}	3.6×10^{-151}	5.0×10^{-890}	2.2×10^{-154}
	Causal Egger	2.1×10^{-109}	4.9×10^{-47}	5.7×10^{-33}	9.2×10^{-98}	3.0×10^{-584}	1.9×10^{-92}
	Causal Joint	6.0×10^{-249}	6.2×10^{-109}	1.1×10^{-55}	8.2×10^{-195}	4.7×10^{-981}	5.6×10^{-160}
	Marginal Multiomics	1.7×10^{-200}	3.2×10^{-123}	6.1×10^{-40}	2.3×10^{-114}	6.2×10^{-1875}	1.3×10^{-191}
	Marginal Pseudobulk	1.3×10^{-137}	4.7×10^{-92}	4.1×10^{-37}	5.3×10^{-97}	1.6×10^{-1172}	1.4×10^{-158}
CD8 T cell	Causal SMR	2.1×10^{-204}	1.8×10^{-72}	2.4×10^{-38}	1.7×10^{-104}	1.6×10^{-614}	8.6×10^{-103}
	Causal IVW	3.1×10^{-156}	1.5×10^{-60}	1.7×10^{-34}	8.7×10^{-100}	1.1×10^{-616}	6.9×10^{-83}
	Causal Egger	3.4×10^{-104}	3.1×10^{-38}	1.2×10^{-28}	6.4×10^{-55}	1.2×10^{-446}	7.1×10^{-48}
	Causal Joint	8.8×10^{-189}	1.4×10^{-76}	3.9×10^{-41}	1.5×10^{-119}	4.6×10^{-687}	4.2×10^{-95}
	Marginal Multiomics	1.7×10^{-150}	3.1×10^{-111}	2.1×10^{-34}	2.9×10^{-75}	7.4×10^{-1466}	4.1×10^{-130}
	Marginal Pseudobulk	1.6×10^{-104}	1.4×10^{-77}	3.6×10^{-17}	9.5×10^{-57}	1.0×10^{-824}	1.7×10^{-77}

For each cell type, method, and reference, an upper-tail hypergeometric test was applied to counts pooled over eligible genes. For each gene, N is the number of background candidate peaks, K the number overlapping the reference, n_{sig} the number of selected significant peaks, and k the selected peaks overlapping the reference. Eligibility requires $N > 0$, $K > 0$, $n_{\text{sig}} > 0$, and $N > K$. The test uses $\sum N, \sum K, \sum n_{\text{sig}}, \sum k$. Entries are hypergeometric p-values, with no additional adjustment across the displayed comparisons specified by the source workbook. Bn, naive B cells; CD4 T cell / CD8 T cell, naive $CD4^+$ / $CD8^+$ T cells. Causal Joint is the Cauchy combination (CCT) of SMR, IVW, and MR-Egger; Marginal Multiomics denotes the WNN metacell correlation and Marginal Pseudobulk the individual-level pseudobulk correlation.

S2 Table. Pooled regulatory-reference fold enrichment across cell types and methods.

Cell type	Method	EnhancerAtlas	FANTOM5	4DGenome	GTEx	HiChIP	PCHiC
Bn	Causal SMR	**3.59**	**6.18**	**5.00**	**2.96**	**4.46**	**2.70**
	Causal IVW	3.10	5.65	4.54	2.71	4.02	2.46
	Causal Egger	2.69	5.59	4.39	2.34	3.89	2.32
	Causal Joint	3.20	5.67	4.42	2.70	4.12	2.56
	Marginal Multiomics	1.98	3.56	2.61	1.70	3.08	1.84
	Marginal Pseudobulk	2.49	5.03	3.76	2.34	3.76	2.38
CD4 T cell	Causal SMR	**3.56**	**4.45**	**3.30**	**3.01**	**4.65**	2.55
	Causal IVW	2.98	4.08	3.12	2.63	4.41	**2.61**
	Causal Egger	2.52	3.46	3.02	2.51	4.17	2.41
	Causal Joint	2.99	4.17	3.20	2.77	4.40	2.56
	Marginal Multiomics	1.84	2.49	1.85	1.68	3.35	1.85
	Marginal Pseudobulk	2.05	3.10	2.30	1.94	3.78	2.16
CD8 T cell	Causal SMR	**3.65**	**4.64**	**3.60**	**2.76**	**5.46**	**2.70**
	Causal IVW	3.13	4.19	3.36	2.69	5.26	2.52
	Causal Egger	2.90	3.84	3.48	2.40	5.08	2.29
	Causal Joint	3.21	4.41	3.46	2.75	5.22	2.53
	Marginal Multiomics	1.90	2.81	2.00	1.68	4.01	1.88
	Marginal Pseudobulk	2.21	3.60	2.11	1.96	4.62	2.07

Fold enrichment is $(\sum k / \sum n_{\text{sig}})/(\sum K / \sum N)$, using the eligible-gene counts defined in S1 Table. Values are rounded to two decimal places. Bold indicates the greatest unrounded enrichment within each cell type and reference column. Contributing genes and selected-set sizes can differ across methods; these descriptive values are not paired tests of method superiority. Bn, naive B cells; CD4 T cell / CD8 T cell, naive $CD4^+$ / $CD8^+$ T cells. Causal Joint is the Cauchy combination (CCT) of SMR, IVW, and MR-Egger; Marginal Multiomics denotes the WNN metacell correlation and Marginal Pseudobulk the individual-level pseudobulk correlation.

S3 Table. Data overview for the three cell types analyzed.

	Bn	CD4 T cell	CD8 T cell
Single-cell data used for the pseudobulk correlation (after quality control)			
Individuals (passing RNA and ATAC cell-count filters)	335	372	346
scRNA-seq cells	160,577	916,643	460,439
scATAC-seq cells	123,580	360,343	210,077
scRNA-seq cells per individual, median (IQR)	436 (315–619)	2,336 (1,655–3,150)	1,200 (763–1,760)
scATAC-seq cells per individual, median (IQR)	348 (236–464)	898 (618–1,278)	510 (320–845)
Genes in the pseudobulk RNA matrix	36,326	36,326	36,326
Peaks in the pseudobulk ATAC matrix	338,036	338,036	338,036
cis-QTL summary statistics			
eQTL sample size	405	406	405
caQTL sample size	385	385	384
Peaks with at least one caQTL at $P < 10^{-8}$	3,449	6,110	3,847
Candidate peak–gene pairs and the Mendelian randomization analysis set			
Candidate gene–peak pairs (All_GenePeak, TSS–peak distance < 1 Mb)	3,814,826	5,146,441	4,505,251
With caQTL and eQTL variants in common (at least 3 shared variants)	46,190	109,961	62,136
Gene and peak present in both pseudobulk and WNN data	40,389	86,204	51,649
After removing pairs with a missing p-value in any base method	40,153	85,782	51,123
Final analysis set (IVW residual variance ≤ 10)	40,087	85,527	50,931
Genes / peaks in the final analysis set	8,149 / 2,951	10,701 / 5,338	9,427 / 3,327
Variants (instruments) per pair in the IVW fit, median (IQR)	22 (10–40)	22 (10–43)	22 (9–41)
Samples for the marginal correlation methods			
Pseudobulk correlation: individuals	335	372	346
WNN correlation: metacells (external PBMC multiome)	80	80	80

Bn, naive B cells (CIMA L4 type Bn_TCL1A); CD4 T cell, naive $CD4^+$ T cells (CD4_Tn_CCR7); CD8 T cell, naive $CD8^+$ T cells (CD8_Tn_CCR7). The Chinese Immune Multi-Omics Atlas (CIMA; Yin et al., Science 2026; OMIX009786) comprises scRNA-seq and scATAC-seq data from 428 Chinese adults annotated into 73 L4 immune cell types. The cis-QTL were mapped within 1 Mb of the gene transcription start site or the peak midpoint, using genotypes of the whole cohort (5,051,397 variants with minor allele frequency > 0.1); the numbers in the table are specific to the three cell types analyzed here. Three sample definitions are used: the QTL sample sizes (summary statistics used by the Mendelian randomization methods), the individuals in the pseudobulk matrices (those passing the RNA and ATAC cell-count filters; pseudobulk correlation), and 80 metacells of an external PBMC multiome data set (WNN correlation). The caQTL and eQTL data come from the same cohort, so the two QTL data sets overlap in samples. Peaks with at least one caQTL at $P < 10^{-8}$ are counted from the significant caQTL files used to select instruments; this is a nominal threshold, not the false-discovery-rate-defined caPeak count of CIMA. Candidate pairs are All_GenePeak pairs with at least 3 variants shared between the clumped caQTL and the eQTL data, whose gene and peak are present in both the pseudobulk and WNN data. Pairs with a missing p-value in any base method (IVW, MR-Egger, SMR, WNN, pseudobulk) or with an IVW residual variance above 10 were removed. IQR, interquartile range.